DBMS Users and Roles of Users

1 Users of DBMS and Roles of DBMS Users

A Database User is defined as a person who interacts with data daily, updating, reading, and modifying the given data. Database users can access and retrieve data from the database through the Database Management System (DBMS) applications and interfaces.

1.1 Actors on the Scene

Many persons are involved in the design, use, and maintenance of a large database with hundreds of users. The people whose jobs involve the day-to-day use of a large database; we call them the "actors on the scene" and people may be called "workers behind the scene" those who work to maintain the database system environment but who are not actively interested in the database itself.

1.2 Types of Database Users

End users are basically those people whose jobs require access to the database for querying, updating, and generating reports. The database primarily exists for their use. Database users are categorized based on their interaction with the database. There are seven types of database users in DBMS. Below mentioned are the types of database users:

1.2.1 Database Administrator

In a database environment, the primary resource is the database itself, and administering these resources is the responsibility of the database administrator (DBA). The DBA is responsible for authorizing access to the database, for coordinating and monitoring its use, and for acquiring software and hardware resources as needed. The DBA is accountable for problems such as breach of security or poor system response time.

1.2.2 Database Designer

Database designers are responsible for identifying the data to be stored in the database and for choosing appropriate structures to represent and store this data. It is the responsibility of database designers to communicate with all prospective database users in order to understand their requirements, and to come up with a design that meets these requirements. Database designers typically interact with each potential group of users and develop views of the database that meet the data and processing requirements of these groups. Each view is then analyzed and integrated with the views of other user groups. The final database design must be capable of supporting the requirements of all user groups.

1.2.3 There are several categories of end-users these are as follows.

A. Casual End users
B. Naive or Parametric End Users
C. Application Programmers
D. Sophisticated End Users
E. Standalone Users
F. Specialised Users

A. Casual End Users

These are the users who occasionally access the database but they require different information each time. They use a sophisticated database query language basically to specify their request and are typically middle or level managers or other occasional browsers. These users learn very few facilities that they may use repeatedly from the multiple facilities provided by DBMS to access it.

B. Naive or Parametric End Users

These are the users who basically make up a sizeable portion of database end-users. The main job function revolves basically around constantly querying and updating the database for this we basically use a standard type of query known as the canned transaction that has been programmed and tested. These users need to learn very little about the facilities provided by the DBMS they basically have to understand the users' interfaces of the standard transaction designed and implemented for their use. The following tasks are basically performed by Naive end-users:

1. The person who is working in the bank will basically tell us the account balance and post-withdrawal and deposits.
2. Reservation clerks for airlines, railways, hotels, and car rental companies basically check availability for a given request and make the reservation.
3. Clerks who are working at receiving end for shipping companies enter the package identified via barcodes and descriptive information through buttons to update a central database of received and in-transit packages.

C. Application Programmers

The application programmers write different application programs and are responsible for developing the user interface. The application programmers are free to develop the user interfaces in any preferred language such as C/C++/Java etc.

D. Sophisticated End Users

These users basically include engineers, scientists, business analytics, and others who thoroughly familiarize themselves with the facilities of the DBMS in order to implement their application to meet their complex requirements. These users try to learn most of the DBMS facilities in order to achieve their complex requirements.

E. Standalone Users

These are those users whose job is basically to maintain personal databases by using a ready-made program package that provides easy-to-use menu-based or graphics-based interfaces, An example is the user of a tax package that basically stores a variety of personal financial data for tax purposes. These users become very proficient in using a specific software package.

F. Specialized Users

The special users are responsible for writing specialized database-related programs and also have the task of creating the actual database as well as implementing technical controls needed to enforce policies and decisions.

1.3 WORKERS BEHIND THE SCENE

There are persons typically not interested in the database itself. We call them the "workers behind the scene," and they include the following categories.

- DBMS system designers and implementers are persons who design and implement the DBMS modules and interfaces as a software package.
- Tool developers include persons who design and implement tools-the software packages that facilitate database system design and use and that help improve performance. Tools are optional packages that are often purchased separately.
- Operators and maintenance personnel are the system administration personnel who are responsible for the actual running and maintenance of the hardware and software environment for the database system.

2 References

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